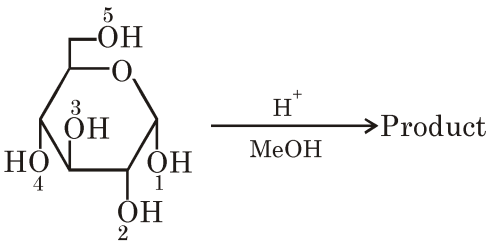
SECTION–I : (i) Only One option correct Type

This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1.



Correct option regarding product

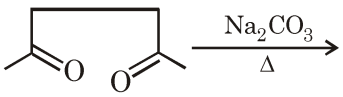
(A) The product gives +ve test with tollens reagent

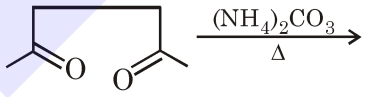
(B) Only $\overset{5}{O}H$ group will be methylated

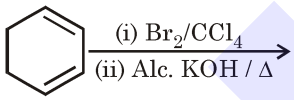
(C) This reaction involves carbanion mechanism

(D) Product when reacts with HIO_4, a new product having multifunctional group is formed. The groups in new product are aldehyde, alcohol and ether.

2. How many reaction give aromatic products ?

(I) 

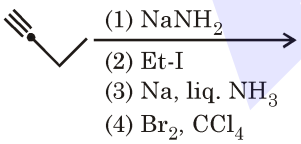
(II) 

(III) 

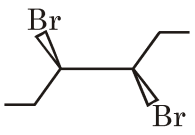
(IV) $CH_3 - C \equiv CH \xrightarrow[\text{Fe tube}]{\text{red hot}}$

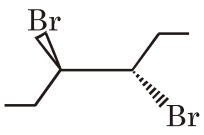
(V) $CH_3 - \underset{\underset{O}{\|}}{C} - CH_3 \xrightarrow[H_2SO_4]{\text{conc.}}$

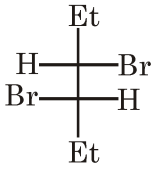
(A) 2 (B) 3 (C) 4 (D) None

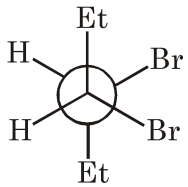
3. 

major product is?

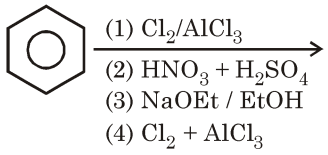
(A) 

(B) 

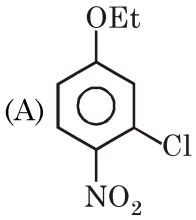
(C) 

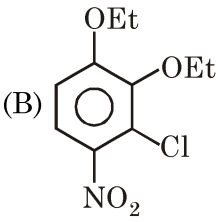
(D) 

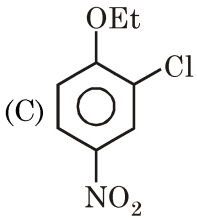
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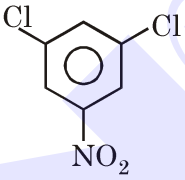
4. 

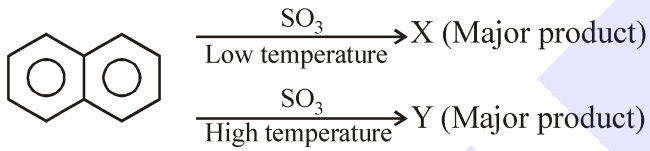
major product is?

(A) 

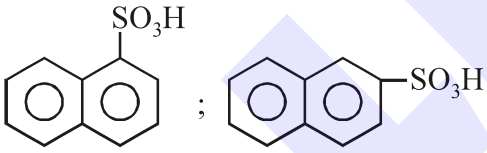
(B) 

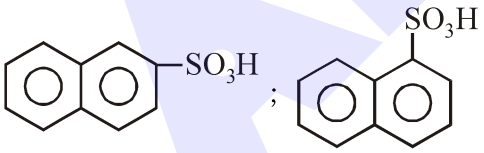
(C) 

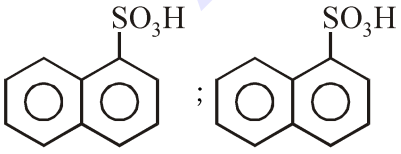
(D) 

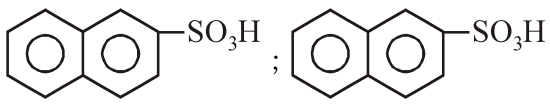
5. 

X and Y are respectively

(A) 

(B) 

(C) 

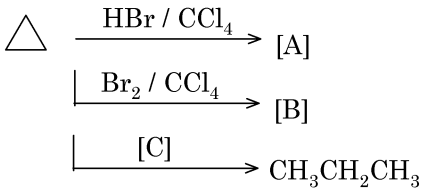
(D) 

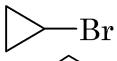
For More Material Join: @JEEAdvanced_2024

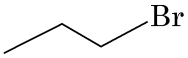
(ii) One or more options correct Type

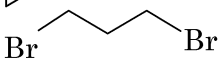
This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

6. In the given reaction identify the unknowns :



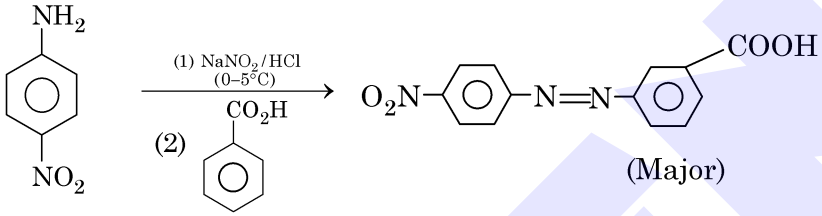
(A) Compound [B] is 

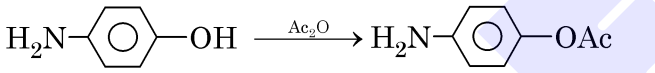
(B) Compound [A] is 

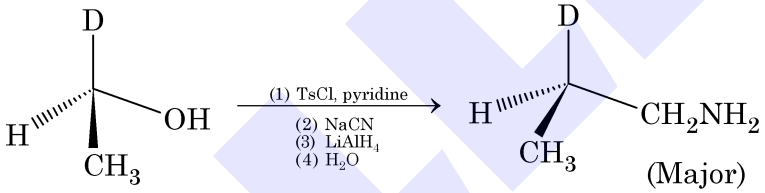
(C) Compound [B] is 

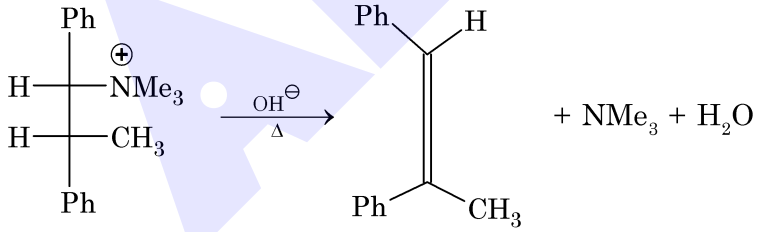
(D) Compound [C] is LAH

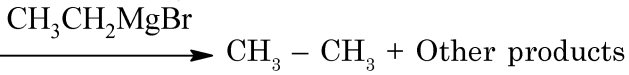
7. Among the following reactions, identify the **INCORRECT** ones.

(A) 

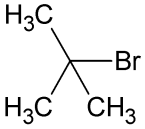
(B) 

(C) 

(D) 

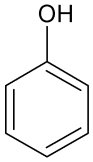
8. **(X)** 

(X) can be?

(A) 

(B) 

(C) $CH_3 - CHO$

(D) 

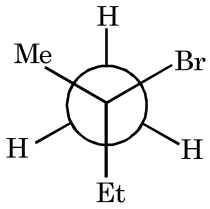
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9. Which of the following statement(s) is/are correct ?

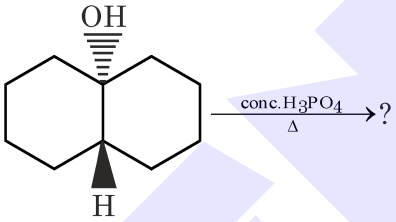
(A) Deoxyribose is present as aldose sugar in nucleic acids.

(B) First member of aldose sugar is aldotriose.

(C)  is more acidic than $CH_2 = CH - CH_2 - CH = CH_2$

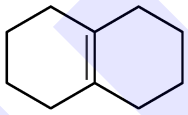
(D) Major alkene product formed by reaction of  and alcoholic KOH/Δ is trans.

10. Consider the reaction given below



Select the correct option(s)

(A) Three products are obtained

(B) The major product is 

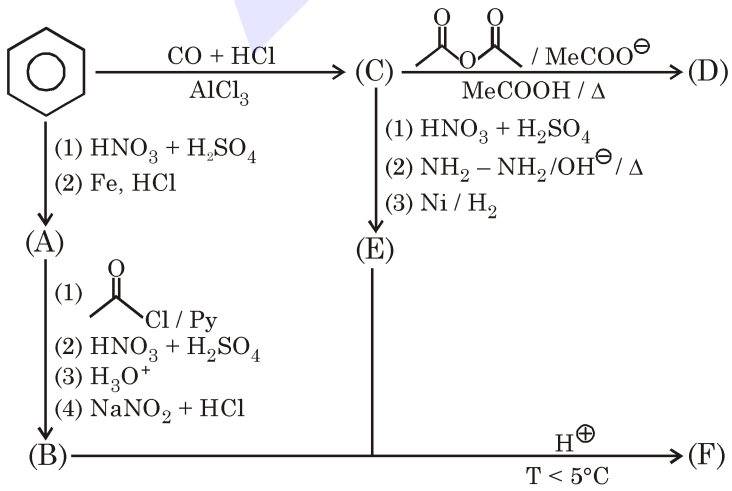
(C) Two products are obtained

(D) Two minor products are enantiomers.

(iii) Paragraph Type

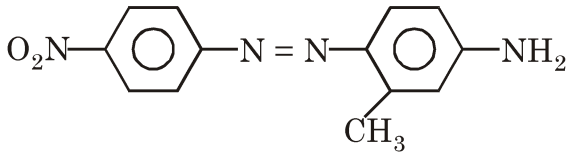
This section contains **01 paragraph1** each describing theory, experiment, data etc. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

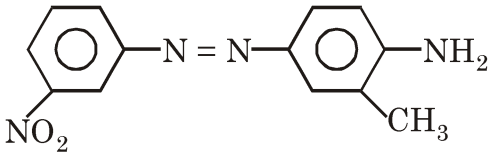
Paragraph for Q. No. 11 to 12

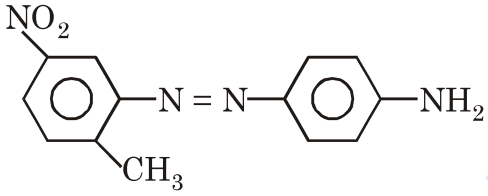


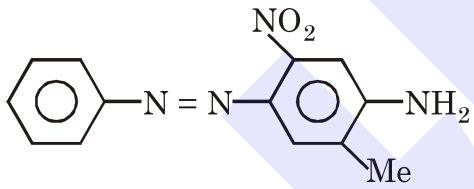
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11. Identify (F) product?

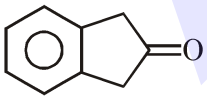
(A) 

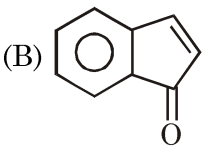
(B) 

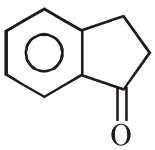
(C) 

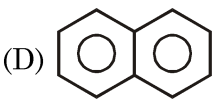
(D) 

12. 'D' product $\xrightarrow[(2)\ SOCl_2]{(1)\ Ni/H_2} \xrightarrow[(3)\ AlCl_3]{}$ major product is?

(A) 

(B) 

(C) 

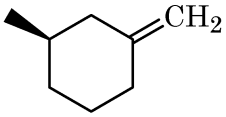
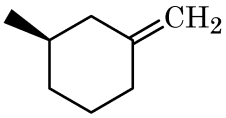
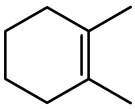
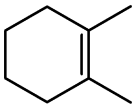
(D) 

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(iv) Matching List Type

This Section contains **2 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

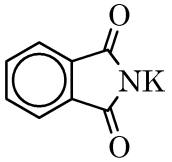
13. Match the List.

List – I (Reaction)		**List – II (Oxidising reagents)**	
(P)	 $\xrightarrow[\text{Reagent}]{\text{Oxidising}}$ 2 organic product	(1)	OsO_4
(Q)	 $\xrightarrow[\text{Reagent}]{\text{Oxidising}}$ 3 organic product	(2)	Bayers reagent
(R)	 $\xrightarrow[\text{Reagent}]{\text{Oxidising}}$ 3 organic product	(3)	(i) O_3, (ii) H_2O
(S)	 $\xrightarrow[\text{Reagent}]{\text{Oxidising}}$ 1 organic product	(4)	(i) $PhCO_3H$, (ii) $H_3O^{\oplus}$

Codes :

	P	**Q**	**R**	**S**
(A)	(1,3)	(3)	(4)	(3)
(B)	(1,2)	(4)	(4)	(1,2,3)
(C)	(3)	(1,3)	(3,4)	(2,3)
(D)	(4)	(2)	(1,2,3,4)	(1,2,3,4)

14. Match the List.

List – I		**List - II**	
(P)	$Mg_2C_3 \xrightarrow{H_2O} A \xrightarrow[NH_4OH]{Cu_2Cl_2} ?$	(1)	Purple colour
(Q)	Hexane $\xrightarrow[\Delta]{Al_2O_3-Cr_2O_3} B \xrightarrow{\text{Oleum}} C \xrightarrow[(ii)H^{\oplus}]{(i)NaOH} D \xrightarrow[FeCl_3]{\text{neutral}} ?$	(2)	Yellow colour
(R)	Ethanol $\xrightarrow{Et_2SO_4} E \xrightarrow[h\upsilon]{air} F \xrightarrow[KCNS]{FeSO_4} ?$	(3)	Red colour
(S)	Ethylpropandioate $\xrightarrow{Na} G \xrightarrow{Br_2} H$  $\xrightarrow[\Delta]{H_3O^{\oplus}} J \xrightarrow{\text{Ninhydrin}} ?$	(4)	No colour

Codes :

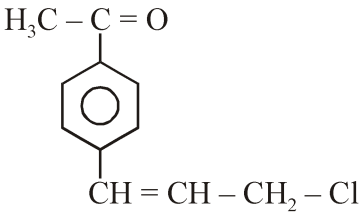
	P	**Q**	**R**	**S**
(A)	(1)	(2)	(3)	(4)
(B)	(3)	(1)	(3)	(1)
(C)	(1)	(3)	(4)	(2)
(D)	(4)	(4)	(1)	(3)

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SECTION-III : (Integer Value Correct Type)

This section contains **05 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

1.



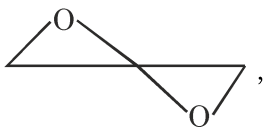
How many reactions are possible for the above compound?

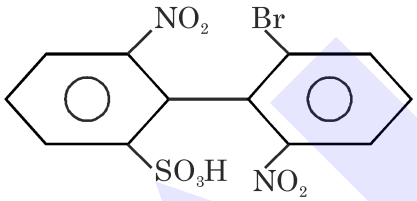
(i) Nucleophilic substitution
(ii) Free radical substituion
(iii) Electrophilic addition
(iv) Nucleophilic addition
(v) Aldol condensation
(vi) Electrophilic substitution

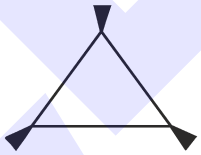
2. Find the total number of substrates more reactive than $CH_3CH_2CH_2CH_2Cl$ with KI in acetone.

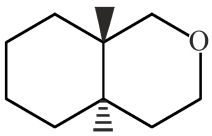
$CH_3Cl, (CH_3)_2CHCl, CH_2=CH-CH_2Cl, PhCH_2Cl, CH_3OCH_2Cl, PhCOCH_2Cl, (CH_3)_3CCl, C_6H_5Cl$

3. Find the total number of optically active compounds







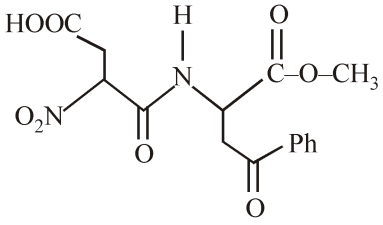


$fac-[Co(gly)_3], [Pt(gly)_2]$

4. How many transition states are formed during the formation of product (major)



5. Total number of functional groups in following compound which can be reduced using $NaBH_4$



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